

Hybrid Feature-Based Detection of AI-Generated Text

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Abstract

1 Introduction

For this project, we explore which linguistic or statistical features are most effective for distinguishing human- vs. AI-generated text. Recent surveys indicate that hybrid approaches, combining statistical and neural methods, often outperform single-modality detectors in robustness and generalization (Wu et al., 2025; Su and Wu, 2024).

The project approach is two-fold. First, a feature analysis is conducted to understand which classic linguistic and statistical features, e.g. sentence complexity, punctuation patterns, lexical diversity, perplexity, and part-of-speech (POS) distributions, are most discriminative. Multiple lightweight classifiers (logistic regression, random forest, XGBoost) are trained on the extracted linguistic/statistical features and their feature importance rankings are compared to identify important/robust signals. Second, the project explores how to make an AI-detector harder to fool in a real world-deployment scenario, specifically, increasing its robustness against adversarial attacks and domain shifts. This is done by combining multiple classification approaches (including neural methods like BERT and statistical methods like Naive Bayes) via majority voting.

RQ: Which linguistic or statistical features are most effective in distinguishing human- vs. AI-generated text?

2 Related Work

3 Data

To ensure our model generalizes across diverse English language sources, domains, and LLMs, and to mitigate overfitting to any single dataset’s biases, we will utilize multiple datasets for training and evaluation. This approach allows for robust testing of features and hybrids in varied contexts, such as essays, Q&A, and long-form texts from different LLMs.

Dataset	Samples	Human (%)	AI (%)
AI Text Detection Pile	1.39M	73.8	26.2
HC3	85K	68.5	31.5
sunilthite	29K	60.1	39.9
DAIGT V2	45K	61.0	39.0
LLM Detect Competition	1.4K	99.8	0.2
AH&AITD	12K	50.0	50.0

Table 1: Dataset size and class distributions

- AI Text Detection Pile (Hugging Face): ~1.3M samples, diverse texts (Reddit, Wikipedia, books) (Artem9k, 2023).
- AH&AITD (Arslan’s Human and AI Text Database): ~12k samples, texts across domains (articles, abstracts, stories, news, reviews) (Akram, 2023).
- LLM - Detect AI Generated Text Dataset (Kaggle sunilthite): >28k essays (Thite, 2023).
- LLM - Detect AI Generated Text (Kaggle Competition): ~10k essays (Kaggle, 2023).
- DAIGT V2 Train Dataset (Kaggle): 48k samples, essays (Deotte, 2023).
- Human ChatGPT Comparison Corpus (HC3): ~24k rows, Q&A from domains (finance, medicine, law, open) (Guo et al., 2023).

3.1 Dataset Balancing

The datasets show varying degrees of class imbalance between the target classes human-written (0) and AI-generated (1). With the extremes being the LLM Detect Competition being 99.8% human-written, while AH&AITD is perfectly balanced. The dataset’s class distribution is shown in Table 1.

The dataset was balanced using a per-source strategy, where each source contributes 50% human and 50% AI-generated samples according to its configured proportion of the total dataset size.

4 Method

The project methodology consists of two efforts to address different aspects of AI text detection: a feature-based analysis and an ensemble-based detector.

4.1 Approach 1: Feature-Based Analysis

The feature-based analysis focuses on understanding which linguistic and statistical features are most informative in classifying human- vs. AI-generated text. For each text in the dataset, a set of features are extracted (section 4.1.1). These features are then used to train three lightweight classifiers: logistic regression and random forest using scikit-learn (Pedregosa et al., 2011), and XGBoost (Chen and Guestrin, 2016).

The dataset was balanced as described in section 3.1, and an 80/20 train-test split was used for all models. Feature importance rankings were retrieved and compared across these three models, giving insights on which linguistic patterns are consistently important for distinguishing human- vs. AI-generated text.

4.1.1 Feature Extraction

For each entry in the dataset, the following features were extracted using the spaCy library (Honninger et al., 2020) with the en_core_web_sm model for all linguistic processing tasks, including tokenization, lemmatization, part-of-speech tagging, and dependency parsing.

Lexical Features Four lexical features were chosen and extracted. Average word length and word length variance to measure vocabulary complexity. Root Type-Token Ratio (RTTR), and hapax legomena ratio to measure lexical diversity.

The Root Type-Token Ratio (RTTR) is a token-type ratio that accounts for document size by normalizing by the square of the document length (Reviriego et al., 2024). It was calculated by lemmatizing the text, counting the unique types and dividing by squared text length. A hapax legomenon refers to a word (or expression) occurring only once in a context, and here measures the proportion of words that appear exactly once in a sample.

Syntactic Features Sentence-level statistics were chosen to measure sentence complexity, and include average sentence length and variance. Parse tree depth features (average, maximum and variance) were extracted to measure syntactic complexity. Parse depth is computed as the maximum number of ancestors for any token in a sentence.

Punctuation Features The ratio of punctuation marks to word count was calculated for comma, period, exclamation mark, and question mark. Additionally, the overall punctuation ratio was included.

Part-of-Speech Features POS tags were extracted using spaCy's coarse-grained POS tags (`token.pos_`) rather than fine-grained tags (`token.tag_`) to capture overall patterns in word class usage, fine grained tags were excluded as they may be too specific for the objective, and yield sparse data. The ratio of each POS category (noun, verb, adjective, adverb, etc.) to total tokens was calculated as well as a noun-to-verb ratio as a derived feature.

Perplexity Perplexity is defined as the inverse test set probability according to the model used for calculating the perplexity score, normalized by the length of the text (Jurafsky and Martin, 2025). It is an evaluation of the model's ability to predict uniformly among the set of tokens in a corpus, or in other words how "surprised" the model is by the text, where a lower perplexity indicates more predictable/less surprising text for the model. Perplexity was calculated using EleutherAI's Pythia 1.4B model (Biderman et al., 2023), an open-source model trained on the Pile dataset (Gao et al., 2020). The model was downloaded from Hugging Face, and run using the transformers library (Wolf et al., 2020). This model was chosen for its balance between computational efficiency and generalization due to its diverse training data, which includes web text, books, and academic sources similar to the datasets of this experiment.

4.2 Approach 2: Ensemble-Based Detector

The ensemble-based detector aims to build a practical, robust detector by combining multiple classifiers via majority voting. The ensemble includes: a BERT classifier; a Naive Bayes classifier; potentially a feature-based model (TODO update after discussion!!). The ensemble works by letting each model making a prediction, then "voting", selecting the majority class. This approach increases robustness against adversarial attacks and domain shifts, as fooling the ensemble would require deceiving a majority of its components simultaneously.

4.3 Evaluation

Classification Performance: All classifiers are evaluated using accuracy, precision, recall, F1-score, and confusion matrices on held-out test sets, for the ensemble both the individual models and the complete ensemble performance is reported.

Feature Importance Analysis: For the feature-based analysis, we extract and compare feature

importances from each model: absolute coefficient values from logistic regression (normalized to sum to 1), Gini importance from random forest, and gain-based importance from XGBoost. By ranking features across models, we identify which linguistic patterns are consistently discriminative regardless of the classification algorithm.

The feature importance analysis compares three classifiers to identify robust signals that generalize across different models (linear vs. tree-based, bagging vs. boosting).

5 Experiment and Results

6 Discussion and Conclusion

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A Example appendix